



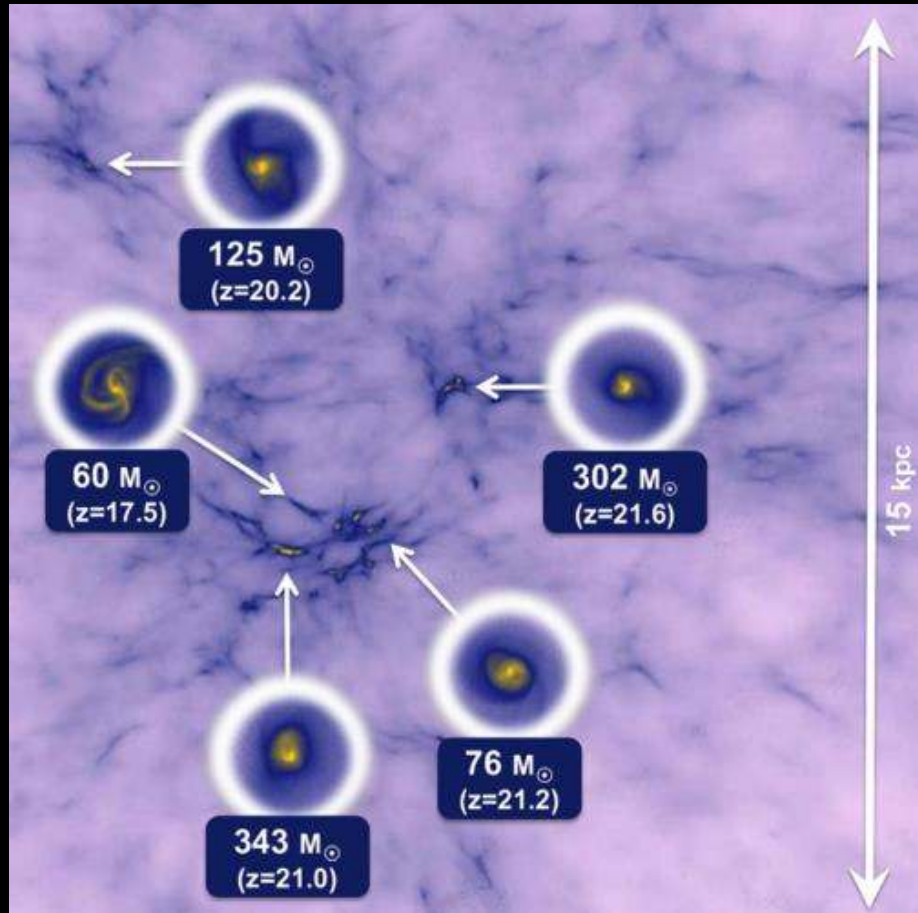
From Population III Stars to Heavy Black Hole Seeds:

Modeling Early Structure Formation with Semi-analytic Models and Simulations

AAS 247 Winter, Phoenix, Arizona

Ryan Hazlett, Eli Visbal, Greg L. Bryan, Mihir Kulkarni

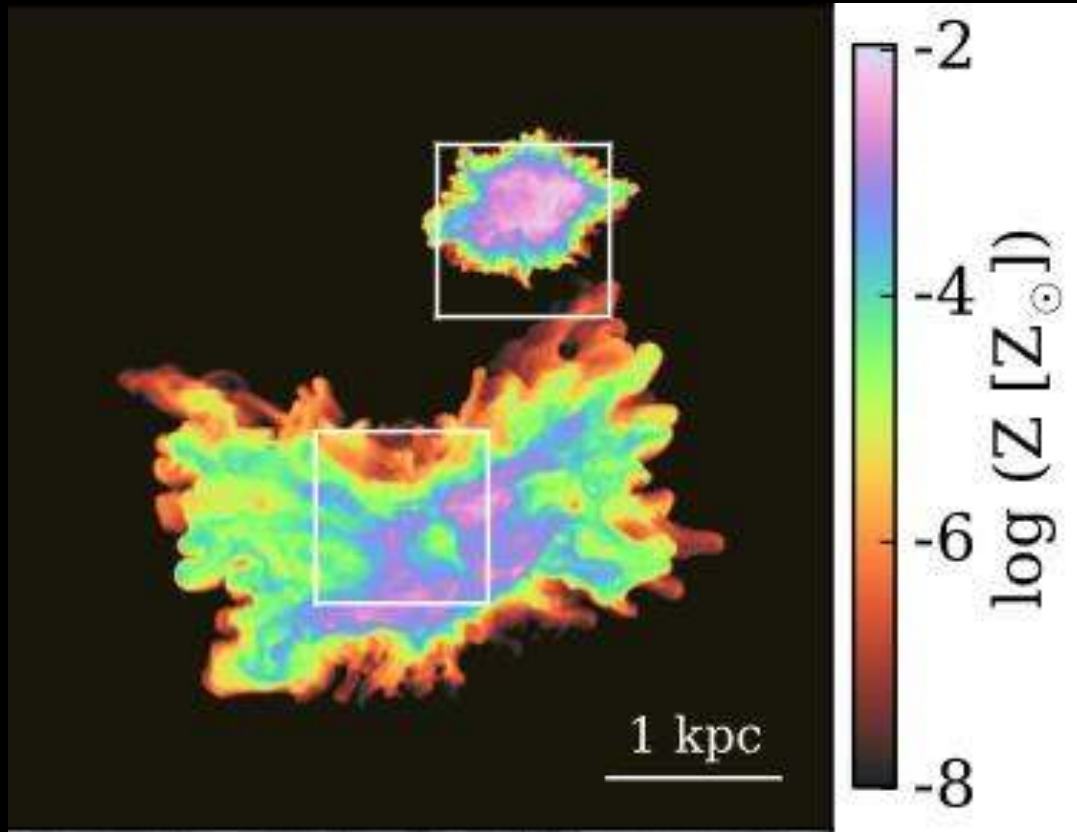
Population III, The First Stars



(Hirano+2014)

- Lack of metals in primordial gas for cooling.
- Simulations indicate massive stars form following a Top-Heavy Initial Mass Function.
(Greif+2011; Hosokawa+2011; Hirano+2014; Susa+2014; Ishigaki+2018; and Klessen+2023 for a recent review)

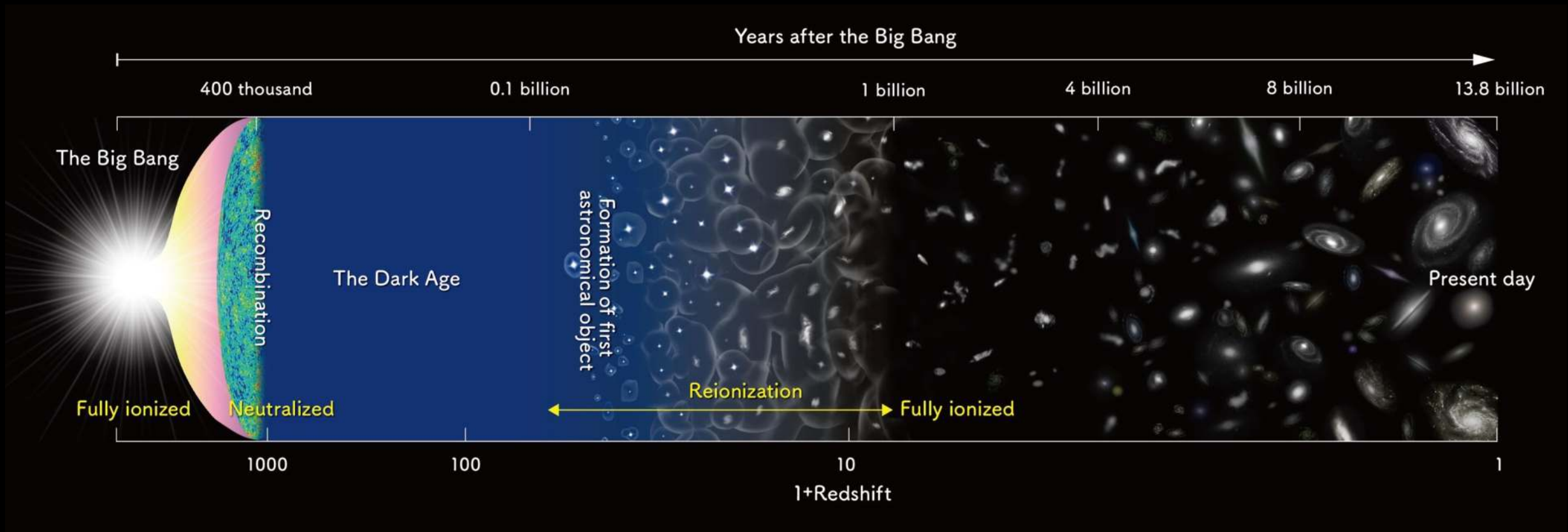
Population II, The First Galaxies



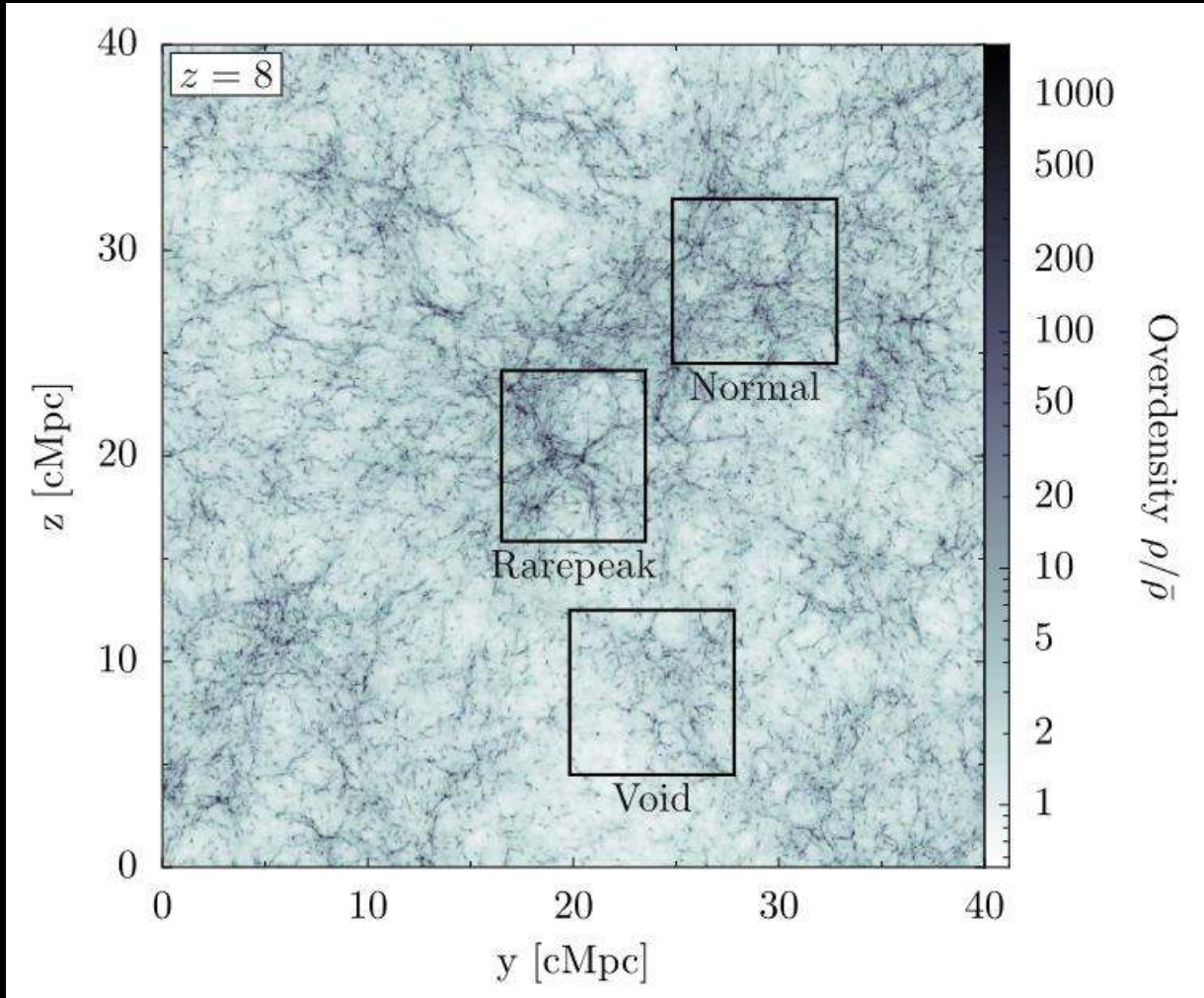
(Smith+2015)

- Pop III supernovae eject metals into nearby ISM and IGM.
- **Metal-enriched** star formation in the host halo, $\sim 10^{-3} - 10^{-6} Z_{\odot}$
- External enrichment leads to **metal-enriched** star formation in nearby halos.

Epoch of Reionization with Simulations



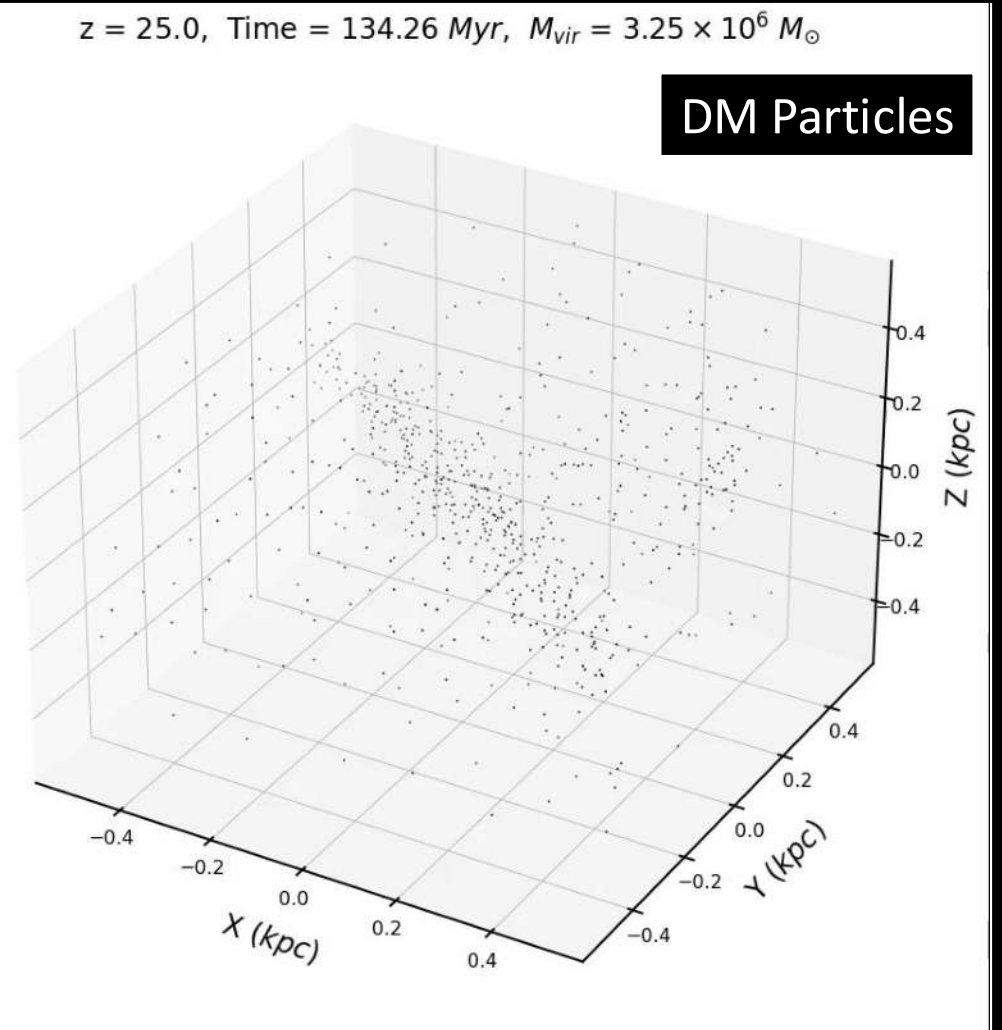
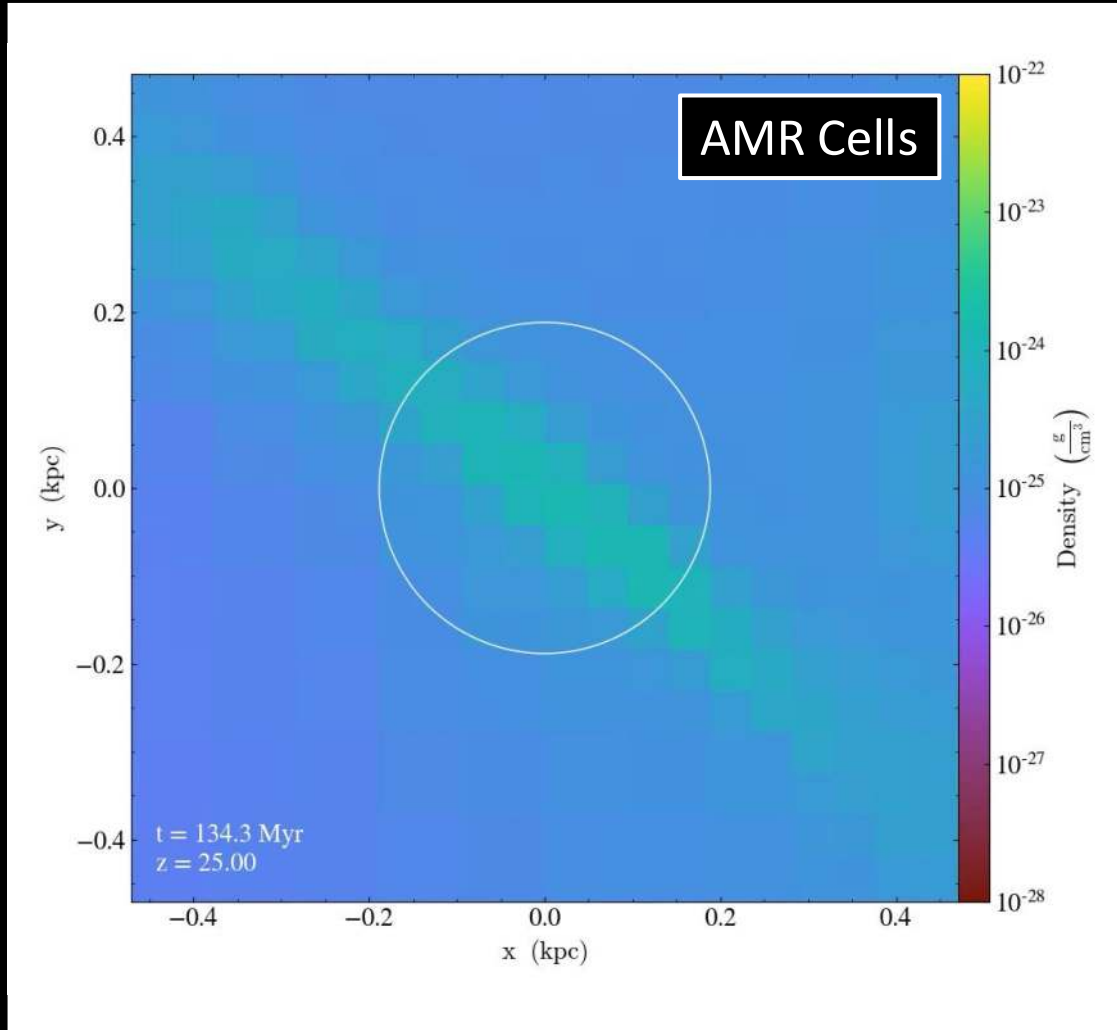
The *Renaissance* Simulations

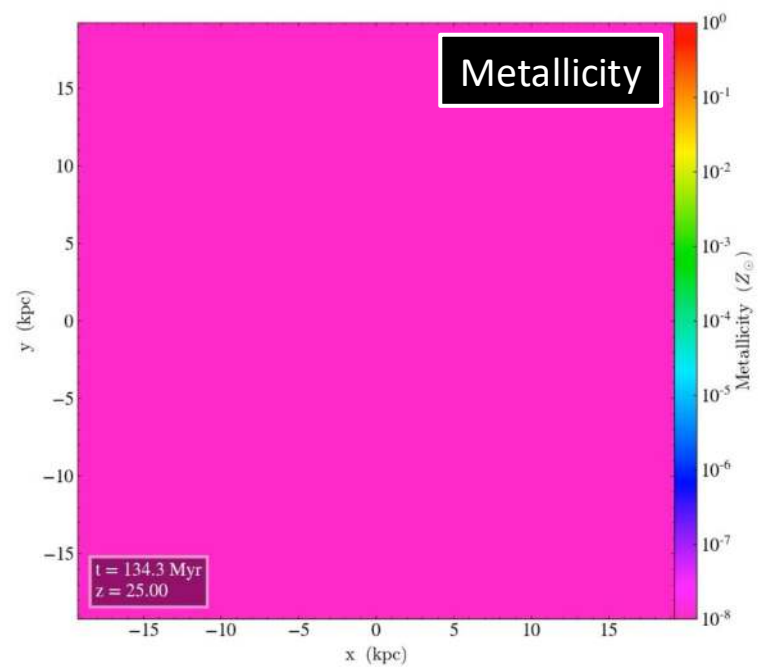
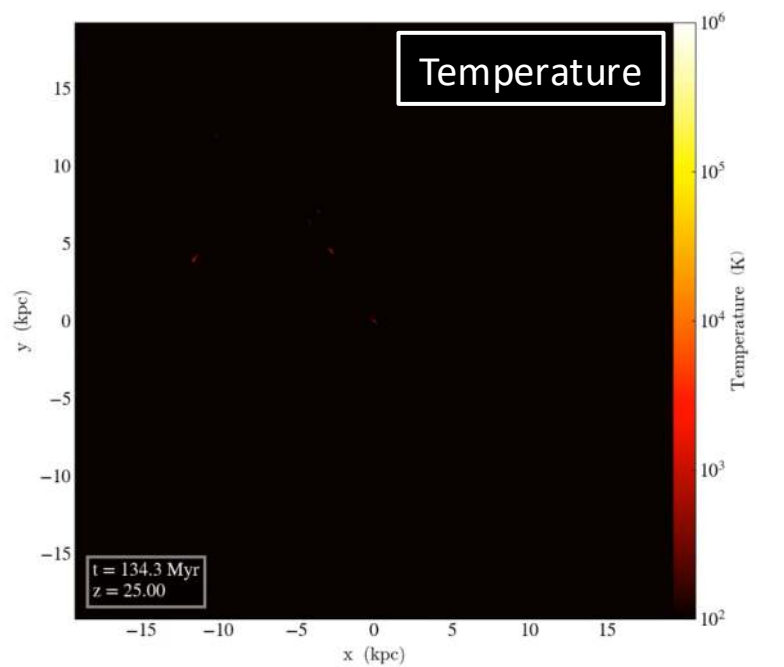
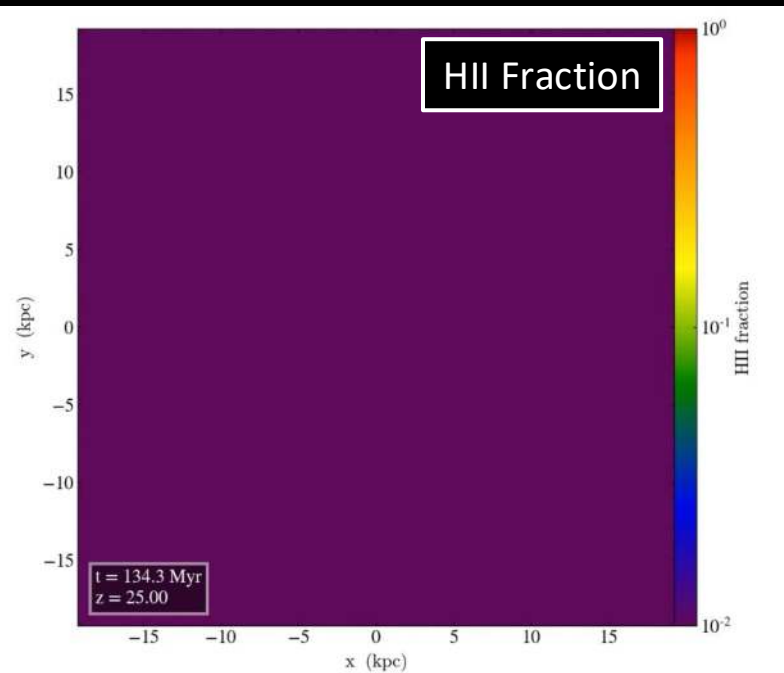
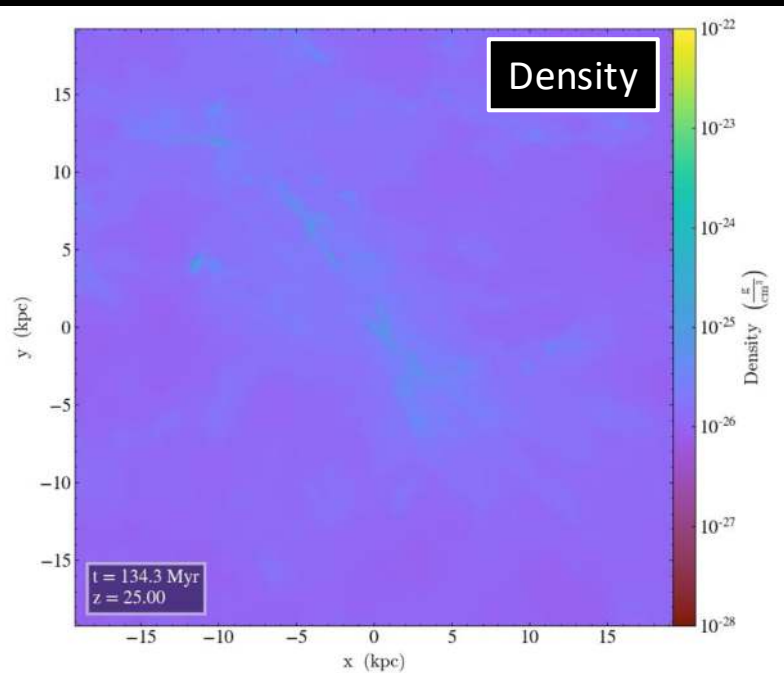


- Cosmological radiative hydrodynamical simulations
- Focused on high redshift galaxy formation and reionization
- Includes metal-free and metal-enriched star formation with feedback
- 3000 early-galaxies with around 7000 Pop III stars

(O'Shea+2015; Xu+2016)

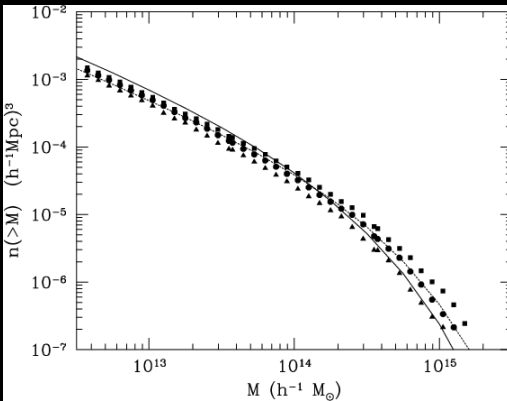
How does *Renaissance* Work?





Advantages of Semi-analytic Modeling

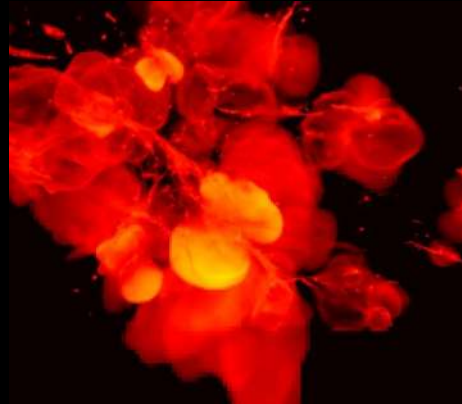
Analytic



Semi-analytic



Radiation Hydro

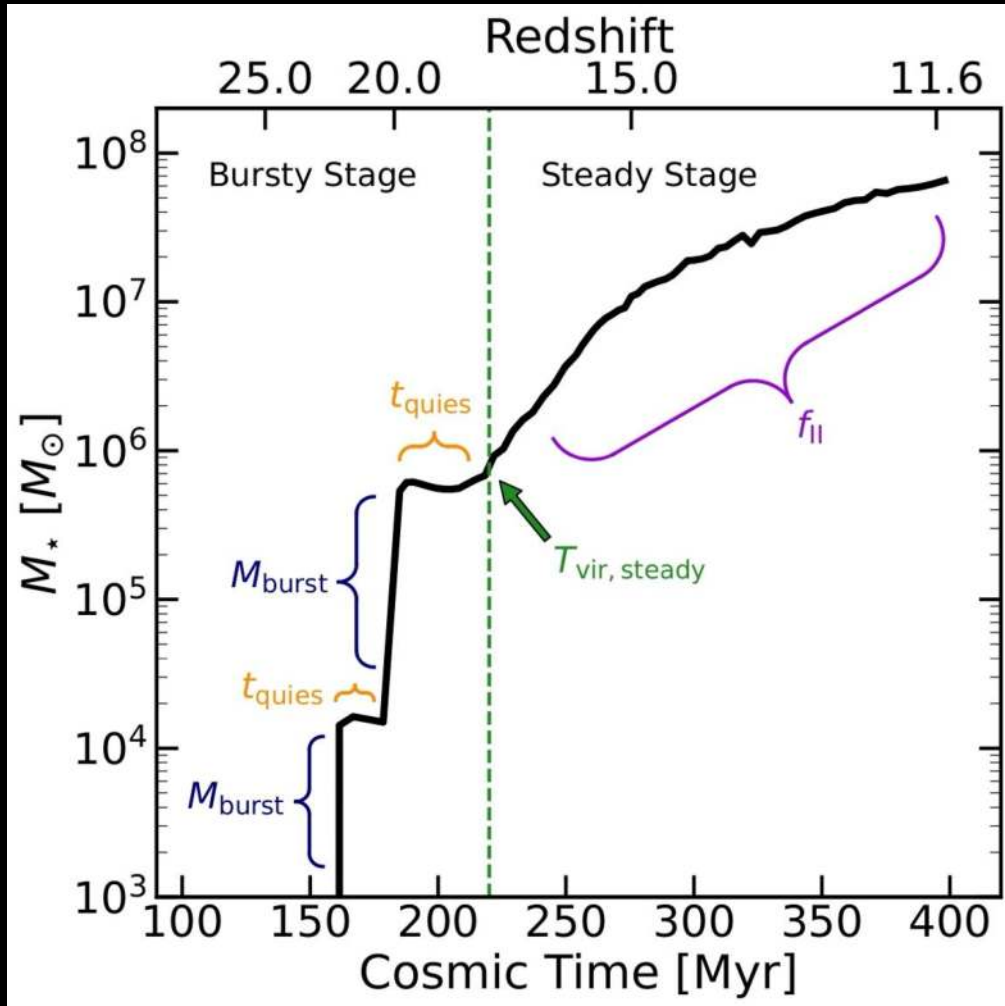


- **Evolution of galaxies using halos from DM-only sims**
- Large volumes like *Renaissance* use 6-million core hours
- Semi-analytic runs complete in a day on a laptop

Realism
Computational Cost & Effort

Ease of parameter studies
Size of cosmological volume possible

Overview of Model Calibration



(Hazlett+2025)

Describe *Renaissance* SFHs with following parameters:

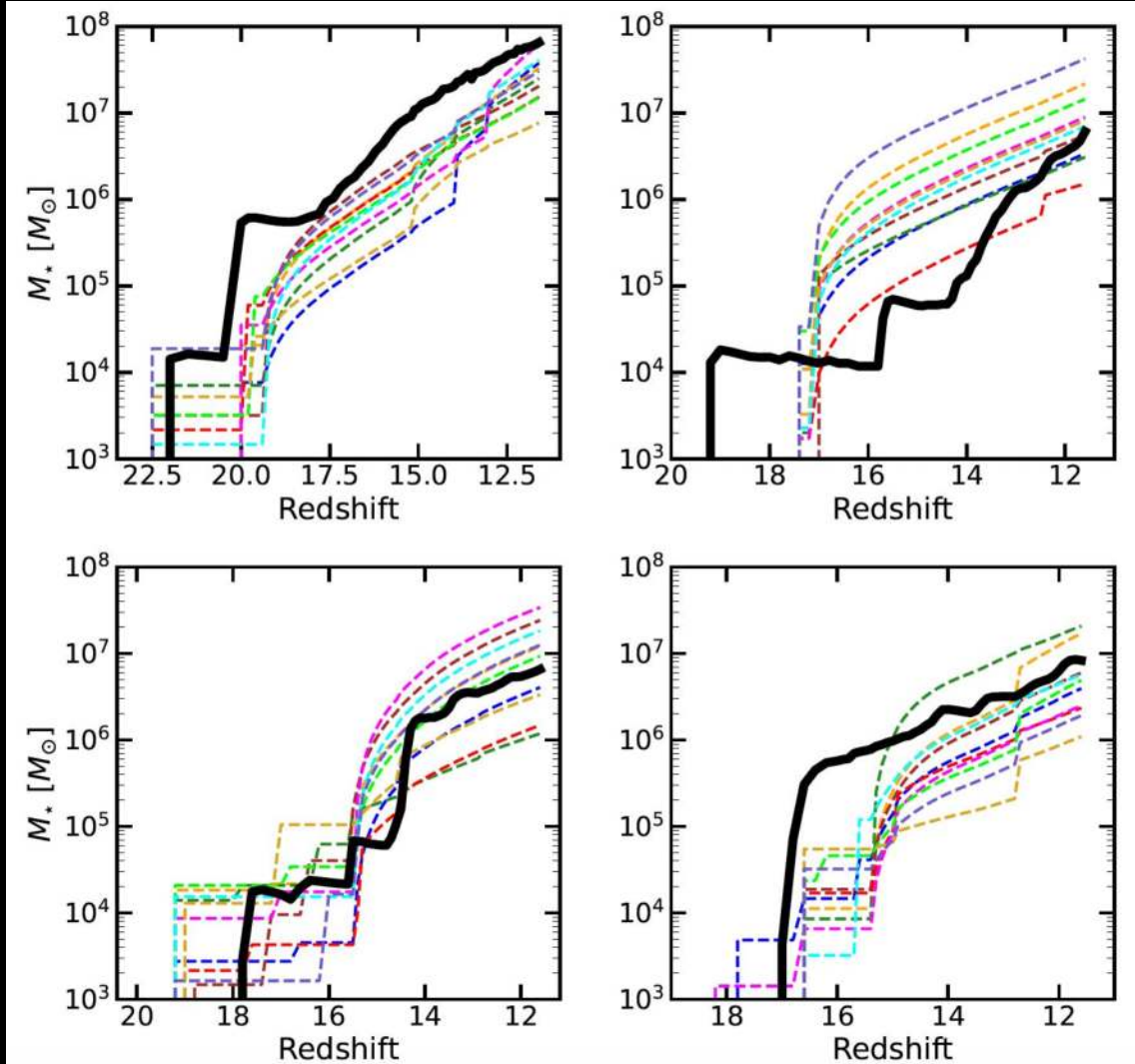
- **Population III**

- M_{crit} : Minimum halo mass for Pop III formation
- M_{PopIII} : Total Pop III mass formed
- t_{delay} : Delay period between Pop III & Pop II

- **Population II**

- M_{burst} : Pop II mass formed in starburst
- t_{quies} : Quiescent period following starburst
- $T_{vir, steady}$: Threshold halo temperature
- f_{II} : Pop II star formation efficiency

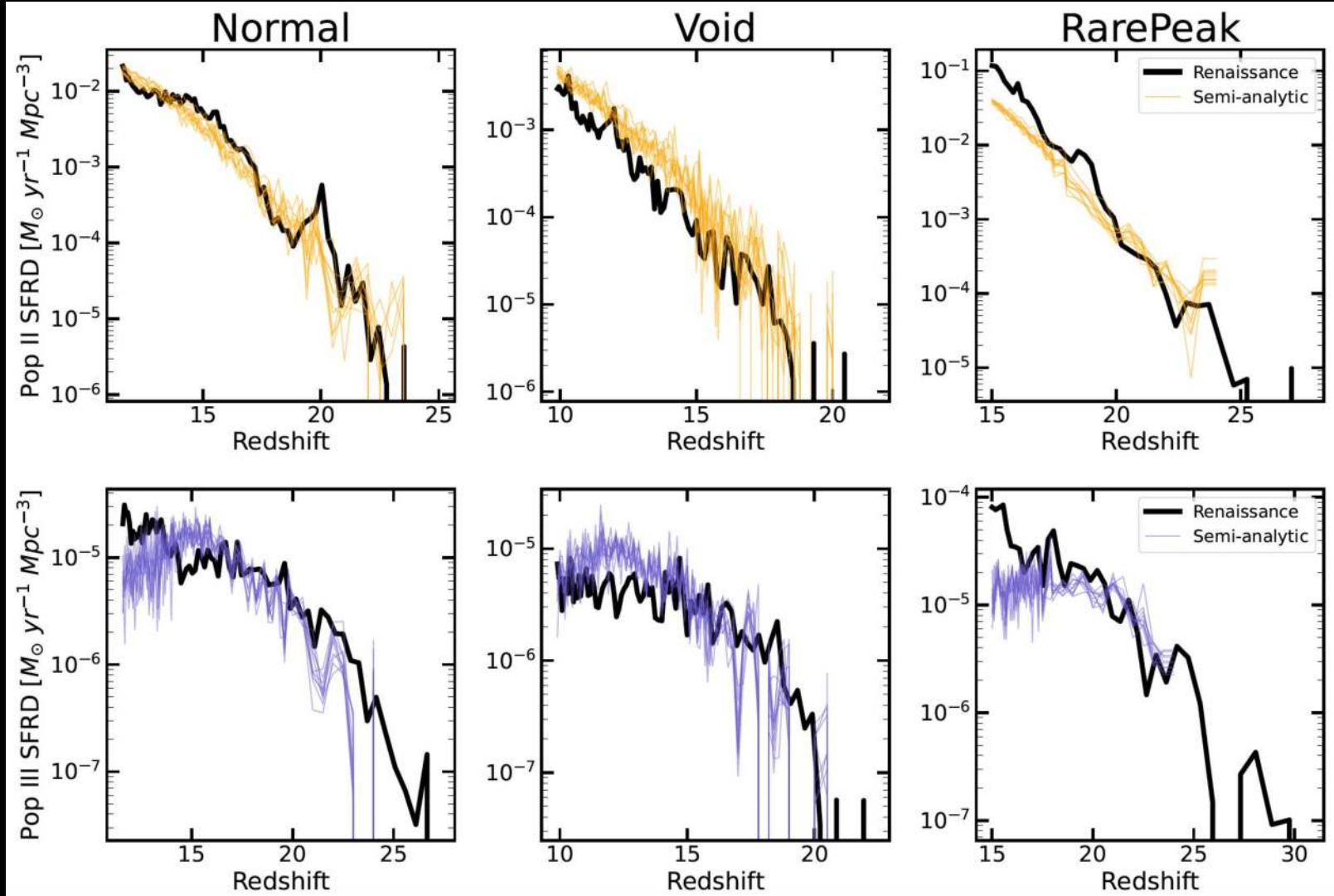
Agreement between individual Pop II SFHs



(Hazlett+2025)

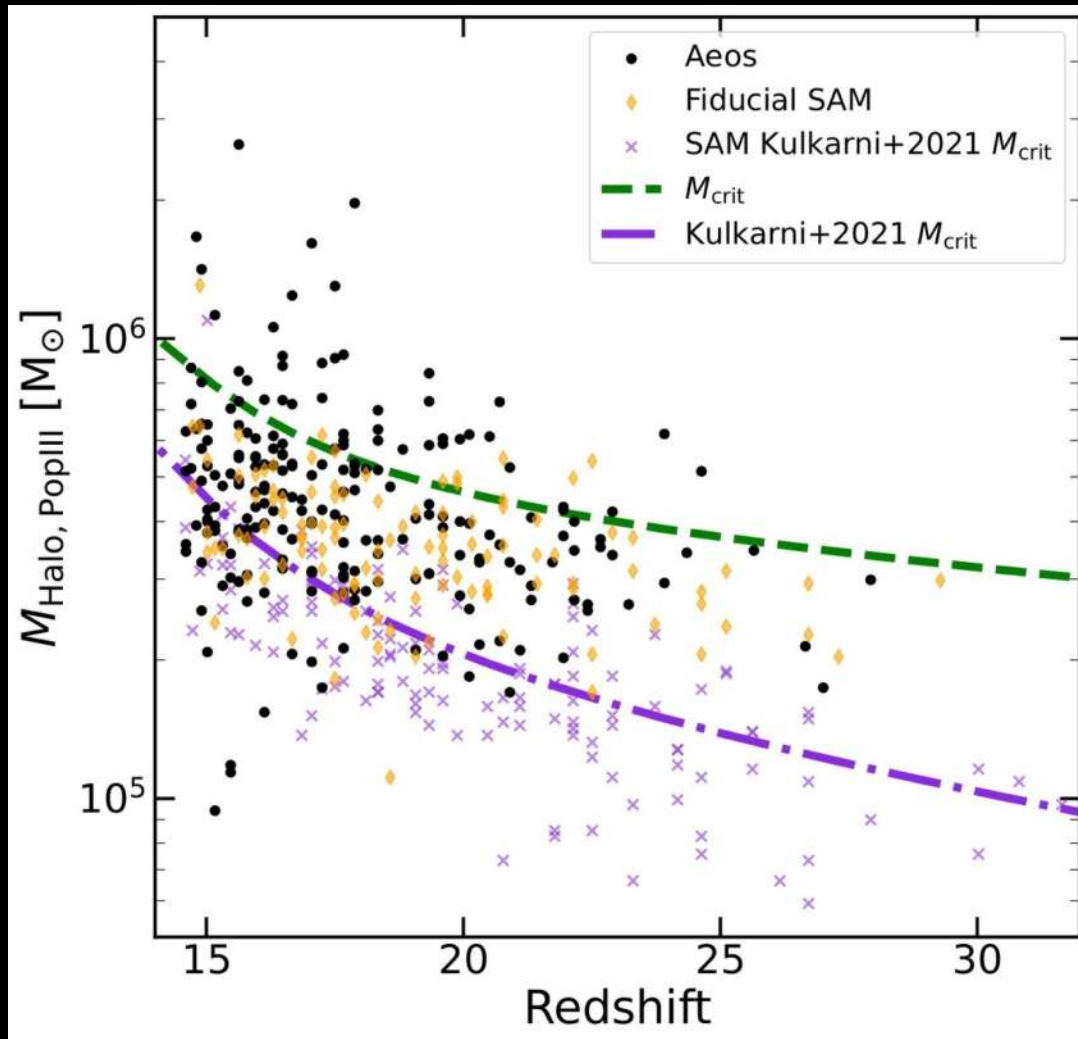
- Gathered a sample of 27 *Renaissance* SFHs
- Identify Pop II starbursts, quiescent periods, and beginning of the steady stage
- Model Pop II SF using relations dependent on halo mass & redshift or a constant SF rate.
- Apply semi-analytic model to *Renaissance* DM halos.

Agreement with *Renaissance* SFRDs



- Tested model on 3 *Renaissance* regions
- Typical agreement within factor of 2 with *Renaissance* Normal
- Possible impact by environmental effects or difference in high mass galaxy evolution

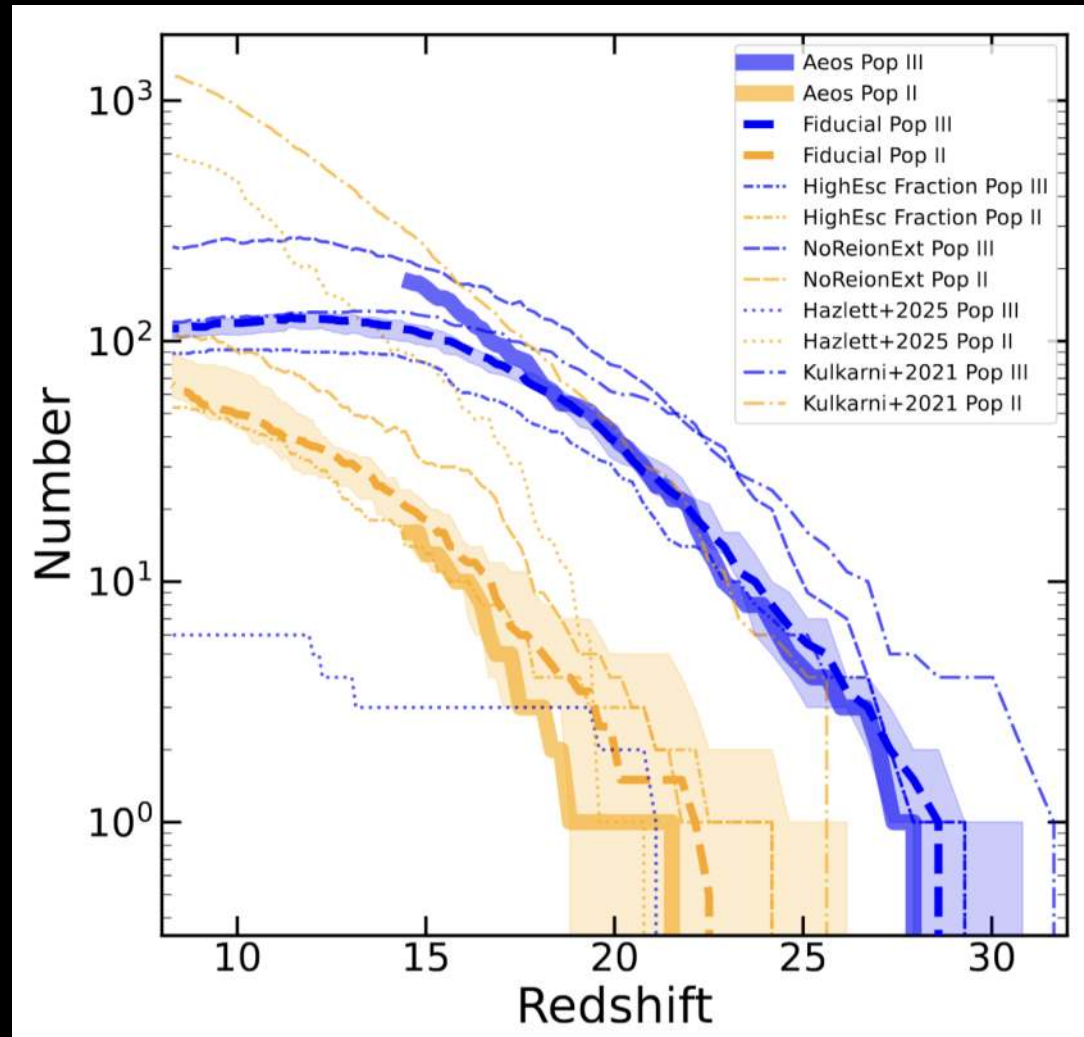
Resolving Pop III formation in Minihalos



(Hazlett+2026 *submitted*)

- *Renaissance* poorly resolves minihalos $\sim 10^{5-6} M_{\odot}$ where Pop III form
- AEOS Simulations (Brauer+2025a):
 - 1 cMpc on a side
 - DM mass resolution $\sim 1840 M_{\odot}$
 - Max spatial resolution ~ 1 pc
- Adding scatter causes systematic excess of low-mass Pop III forming halos

Now have SAM calibrated to Pop III & Pop II



(Hazlett+2026 *submitted*)

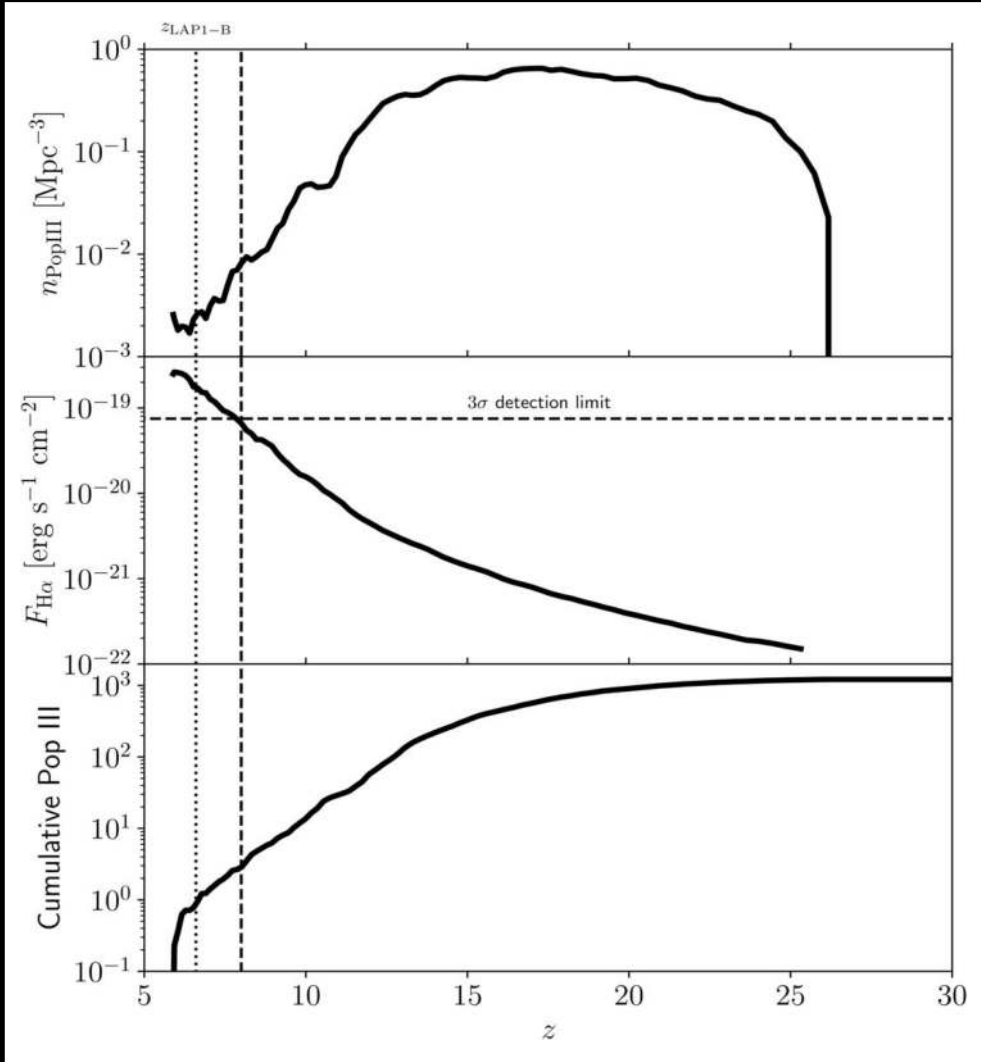
LAP1-B is the First Observed System Consistent with Theoretical Predictions for Population III Stars

see Visbal+2025; arXiv: 2508.03842

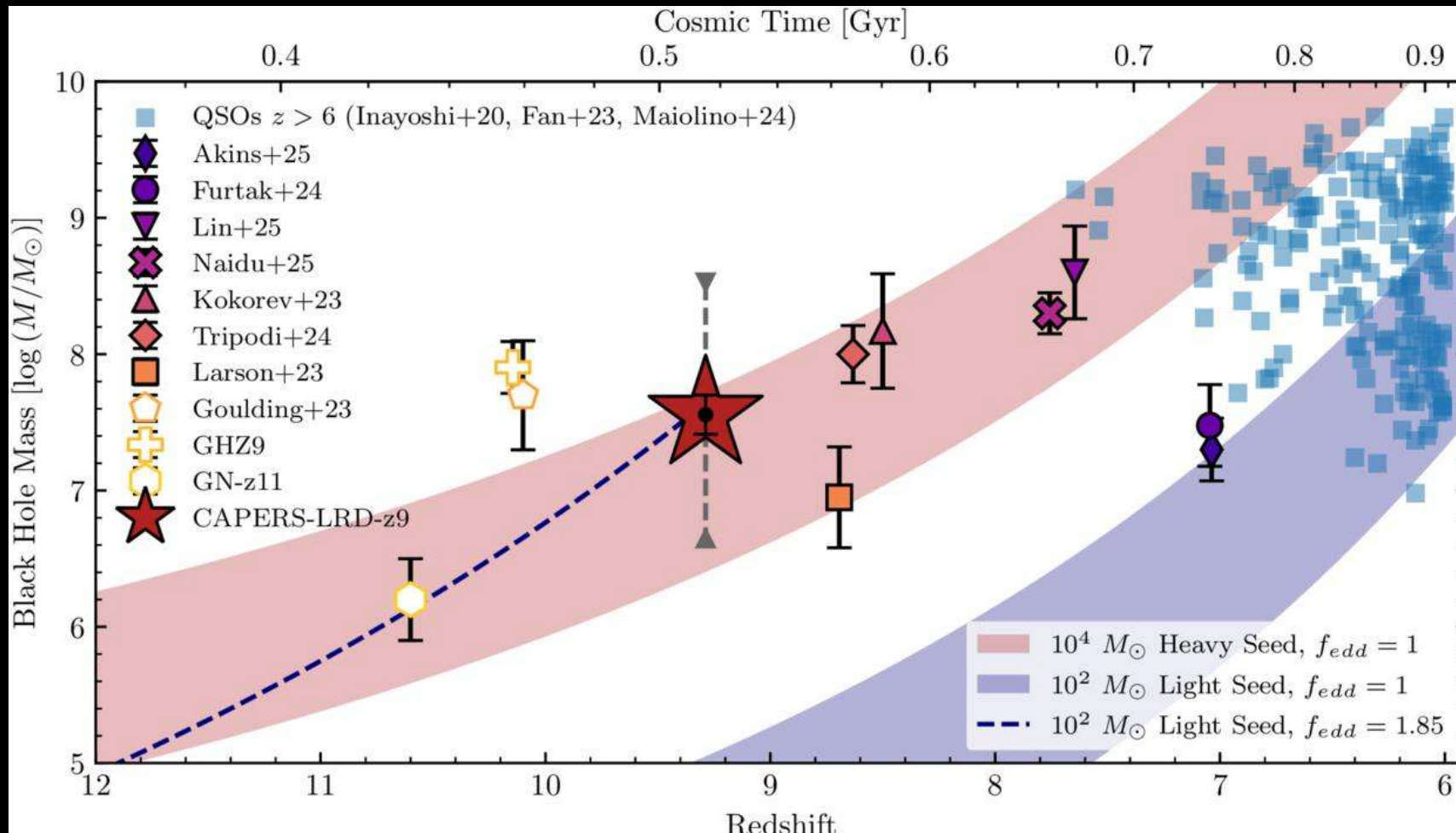
- JWST observations of Pop III candidate LAP1-B gravitationally lensed by galaxy cluster MACS J0416 (Vanzella+2023; Nakajima+2025)

Agrees with 3 key theoretical predictions:

1. Formation in extremely low-metallicity halos with $T_{vir} \sim 10^4$
 2. A top-heavy initial mass function
 3. Formation of low-mass clusters with $\sim$ a few $\times 1000 M_{\odot}$ in massive Pop III stars.
- Our model predicts about one observable Pop III galaxy like LAP1-B in the range $z = 6-7$



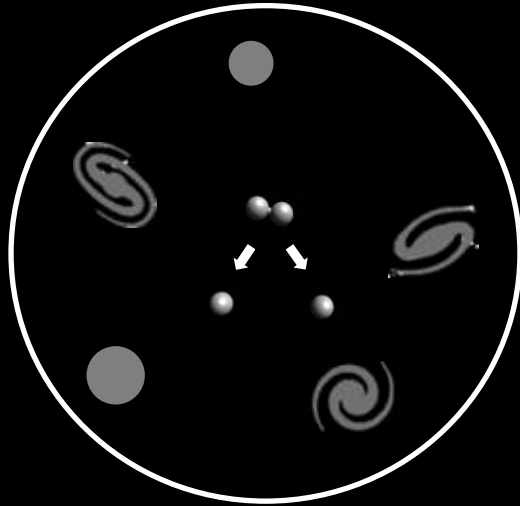
Explaining Massive Black Holes at High-z



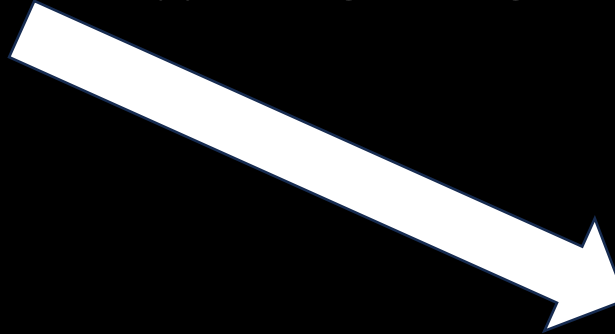
- Shaded curves are seed black holes from the first stars growing at Eddington rate.
- Growing continuously at Eddington rate is difficult.
- Potential explanation: larger mass seed black holes in the early universe.

How to form Massive Seed Black Holes

Lyman-Werner Background

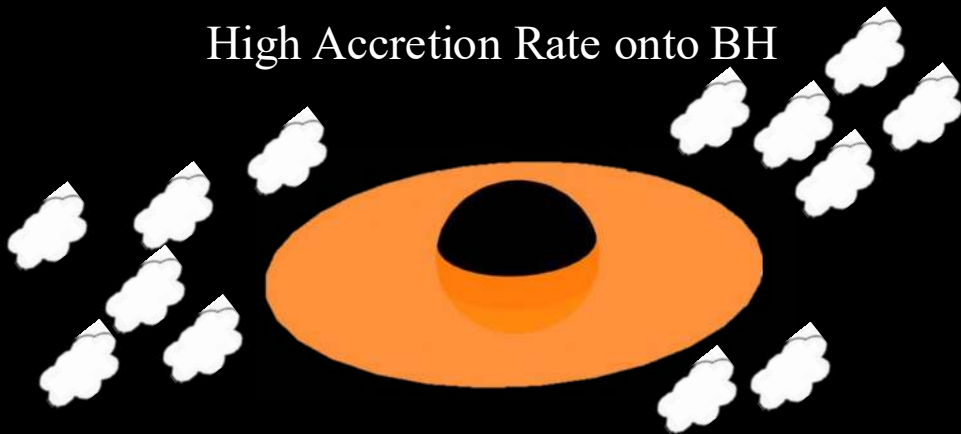


UV photons dissociate H_2
suppressing cooling



Intermediate Mass Seeds
 $10^3 - 10^4 M_{\odot}$

High Accretion Rate onto BH

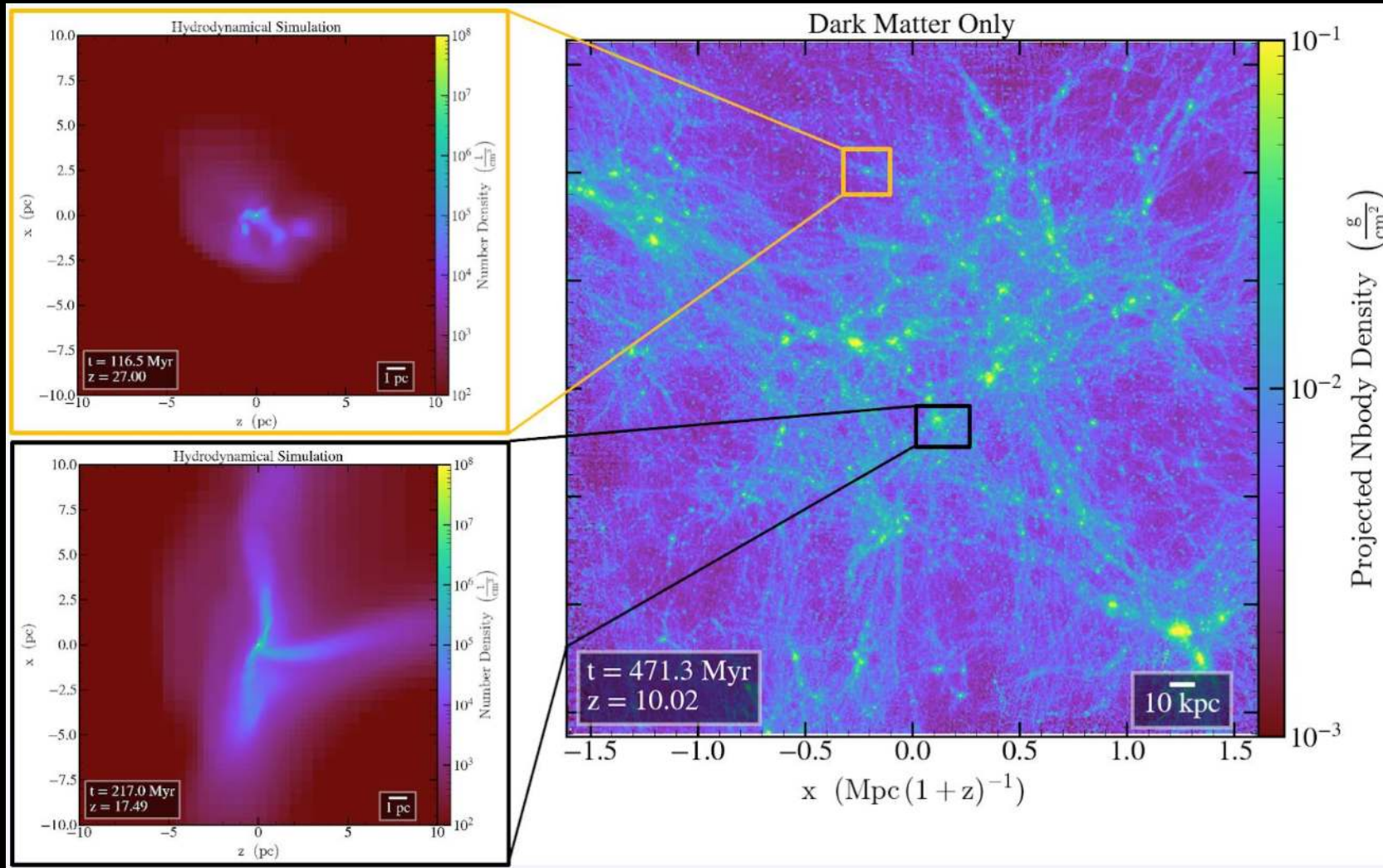


High accretion dynamically
heats the gas



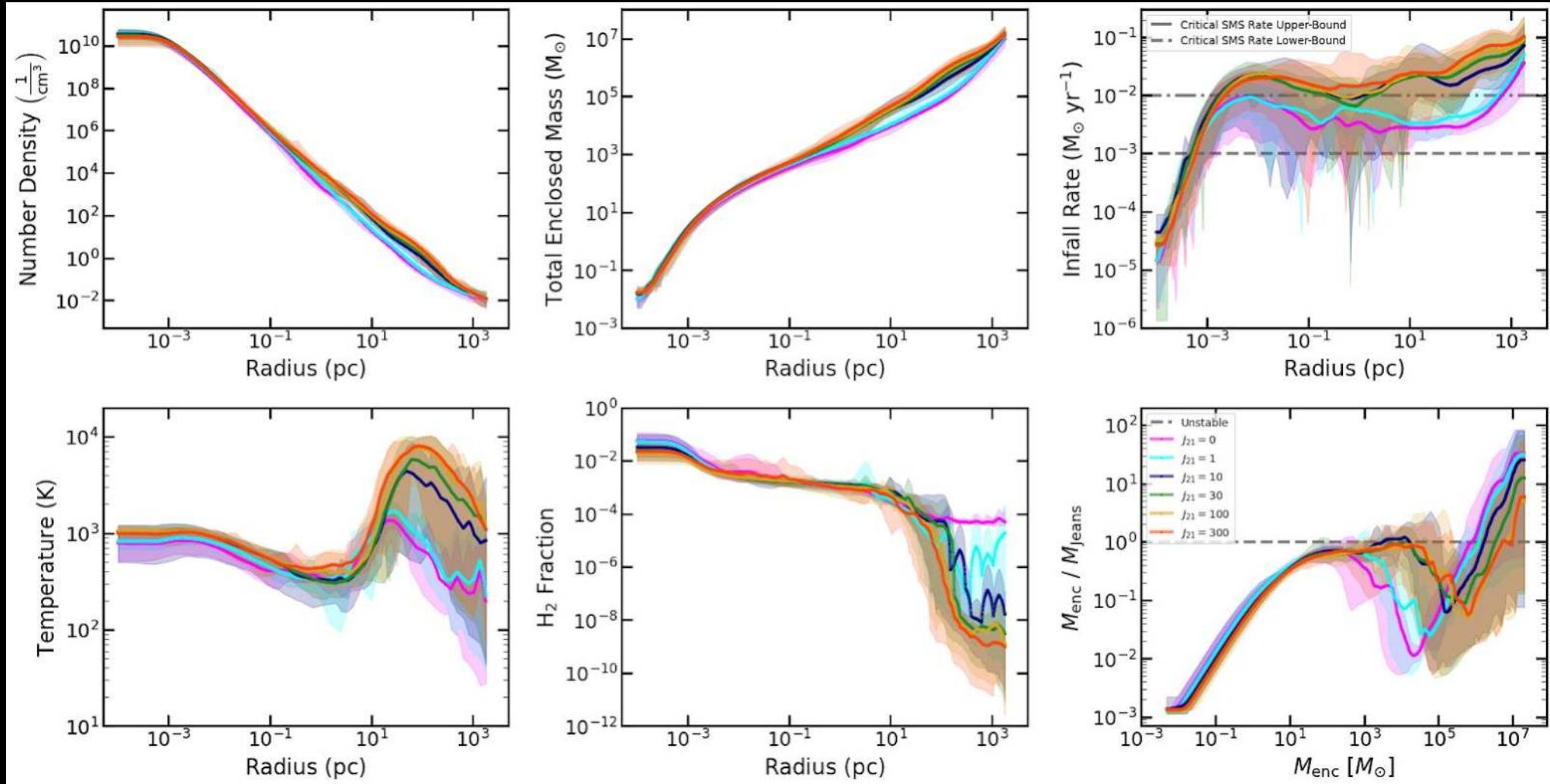
Supermassive Seeds
 $\geq 10^5 M_{\odot}$

Simulate Formation of Seed Black Holes



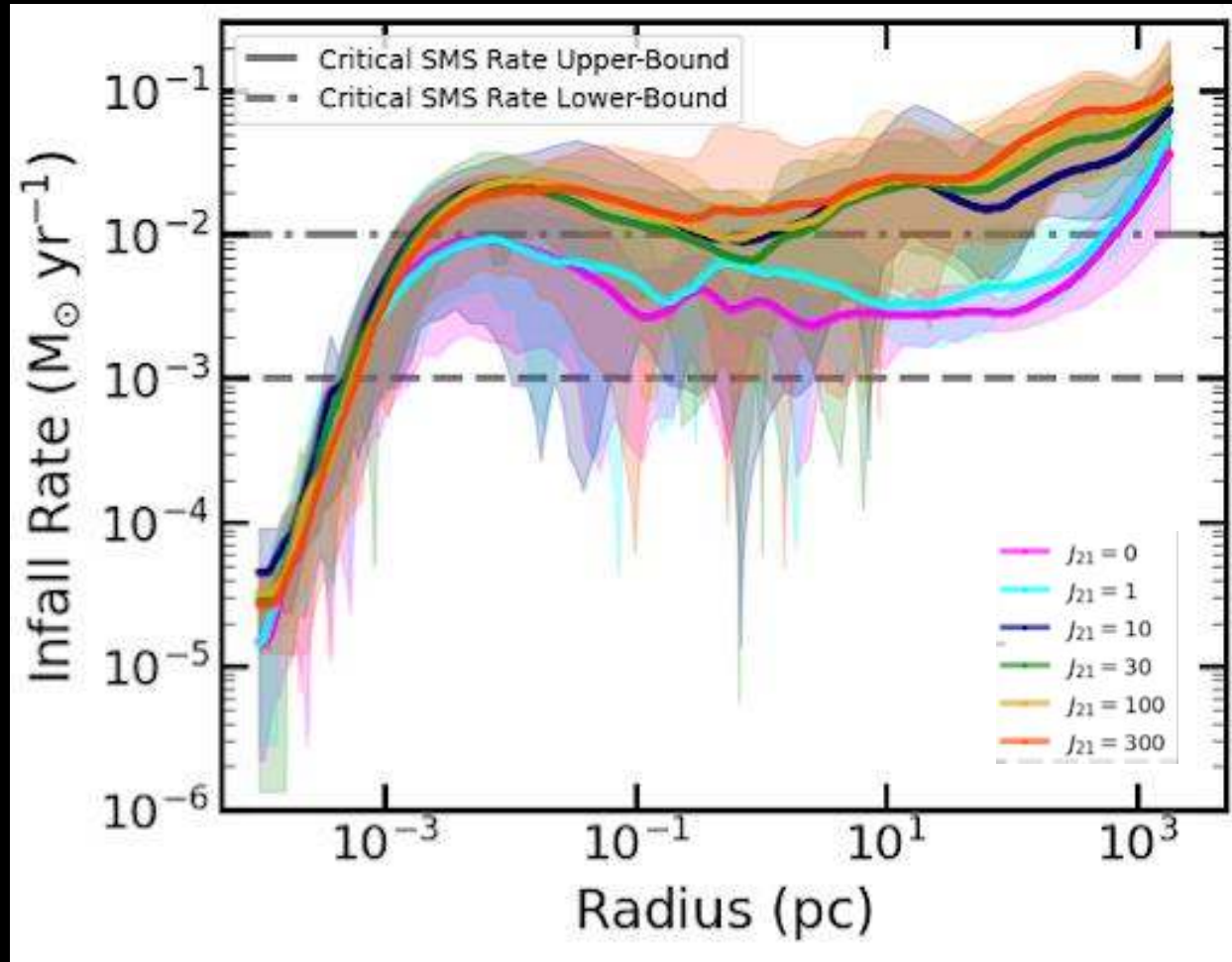
1. Simulate large DM-only volume with adaptive mesh refinement code ENZO (Bryan+2014).
2. Sample of pristine halos with varying J_{21} , halo assembly rates, redshift.
3. High-resolution zoom-in sims with baryons to the onset of Pop III formation.
4. Estimate Pop III stellar mass & BH seed mass.

~50 Zoom-in Simulations at start of collapse



(Hazlett in prep.)

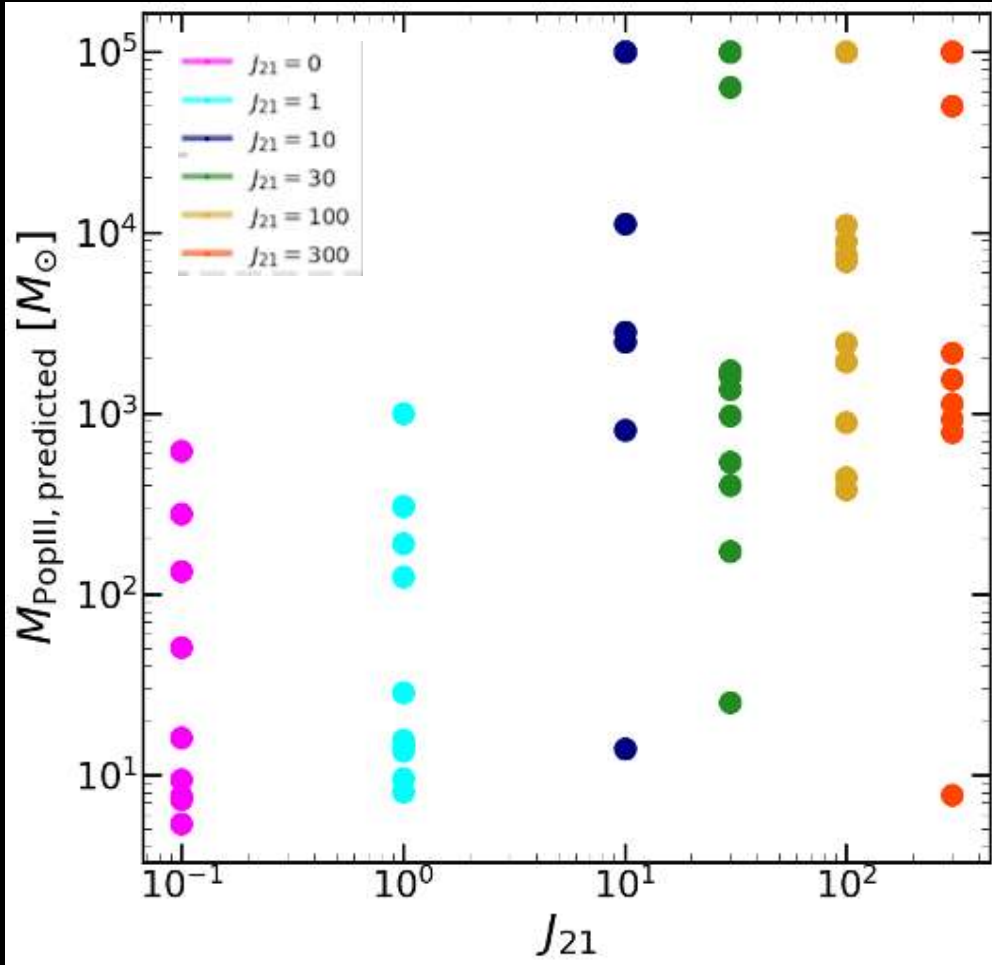
Elevated Infall Rates for $J_{21} \geq 10$



(Hazlett in prep.)

- Compute inflow rates from spherically averaged radial profiles.
- Assume accretion proceeds until inflow drops below critical threshold ($6.7 \times 10^{-3} M_{\odot} \text{ yr}^{-1}$).
- Validate against sink particle simulations (Suazo+2019), finding agreement within factor of ~ 3 .
- $J_{21} \geq 10$ inflow rates exceed threshold, suggests heavy BH seed formation.

Massive BH Seeds more common than previously believed



Note: $J_{21} = 0$ is set to 10^{-1} in this plot &
 J_{21} is $10^{-21} \text{ erg cm}^{-2} \text{ s}^{-1} \text{ Hz}^{-1} \text{ sr}^{-1}$

Findings:

- Moderate LW fluxes ($J_{21} \sim 10$) sufficient to sustain formation of supermassive stars.
- Far below traditionally assumed critical threshold $J_{\text{crit}} \sim 10^4 J_{21}$ (Shang+2010).
- Dynamical heating is less effective compared to LW flux in determining outcomes.
- Heavy BH seeds orders of magnitude more common than in rare LW environments.
- Suggests SMBH growth at high- z is less reliant on extremely rare conditions.

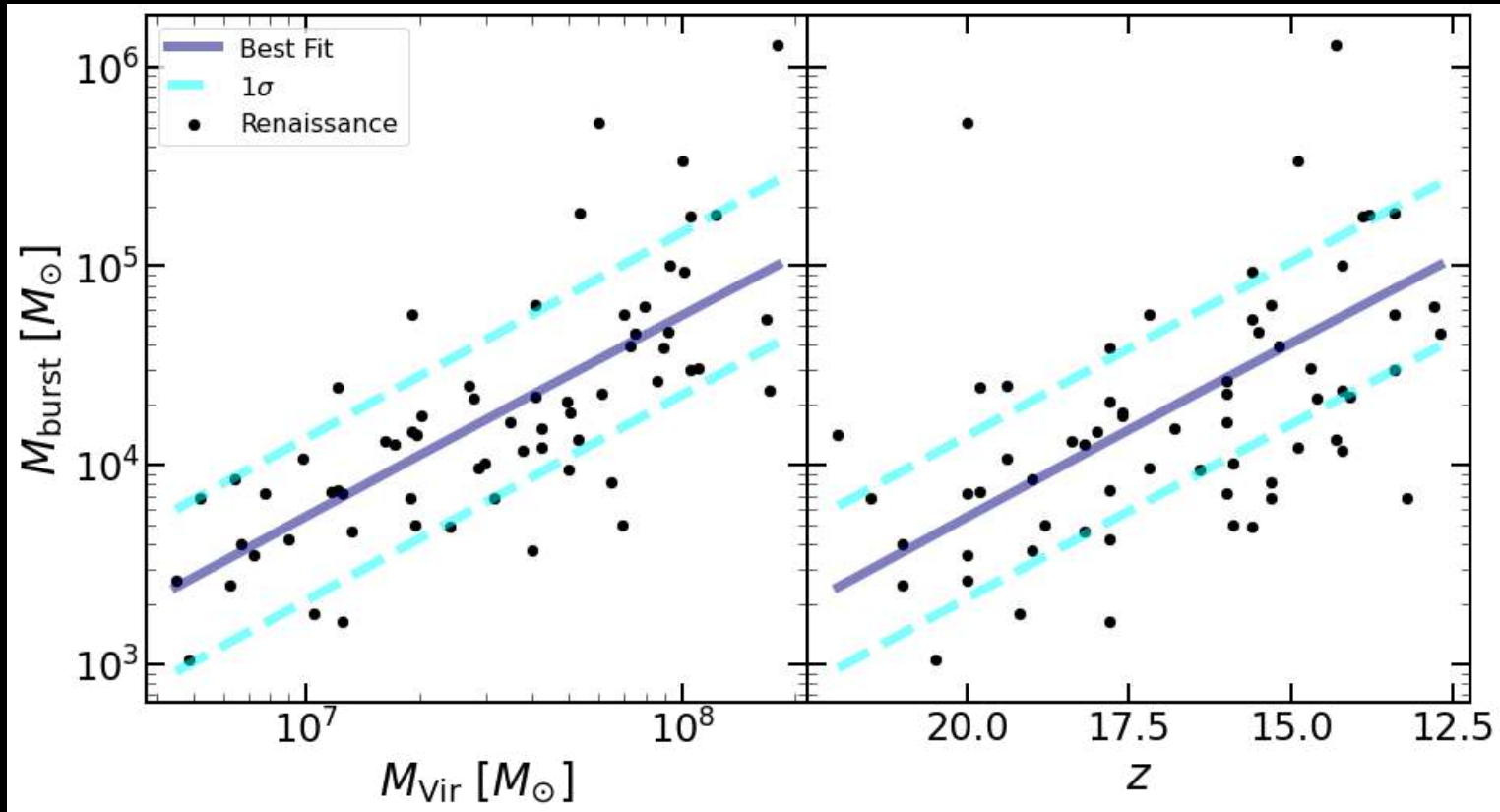
Takeaways

- We have calibrated a semi-analytic model to Pop III & Pop II star formation in two cosmological hydrodynamical simulations
- Our models can be & already are being used to interpret high-z observations of Pop III candidates
- High resolution zoom-in simulations suggest moderate LW fluxes ($J_{21} \sim 10$) are sufficient to form massive BH seeds



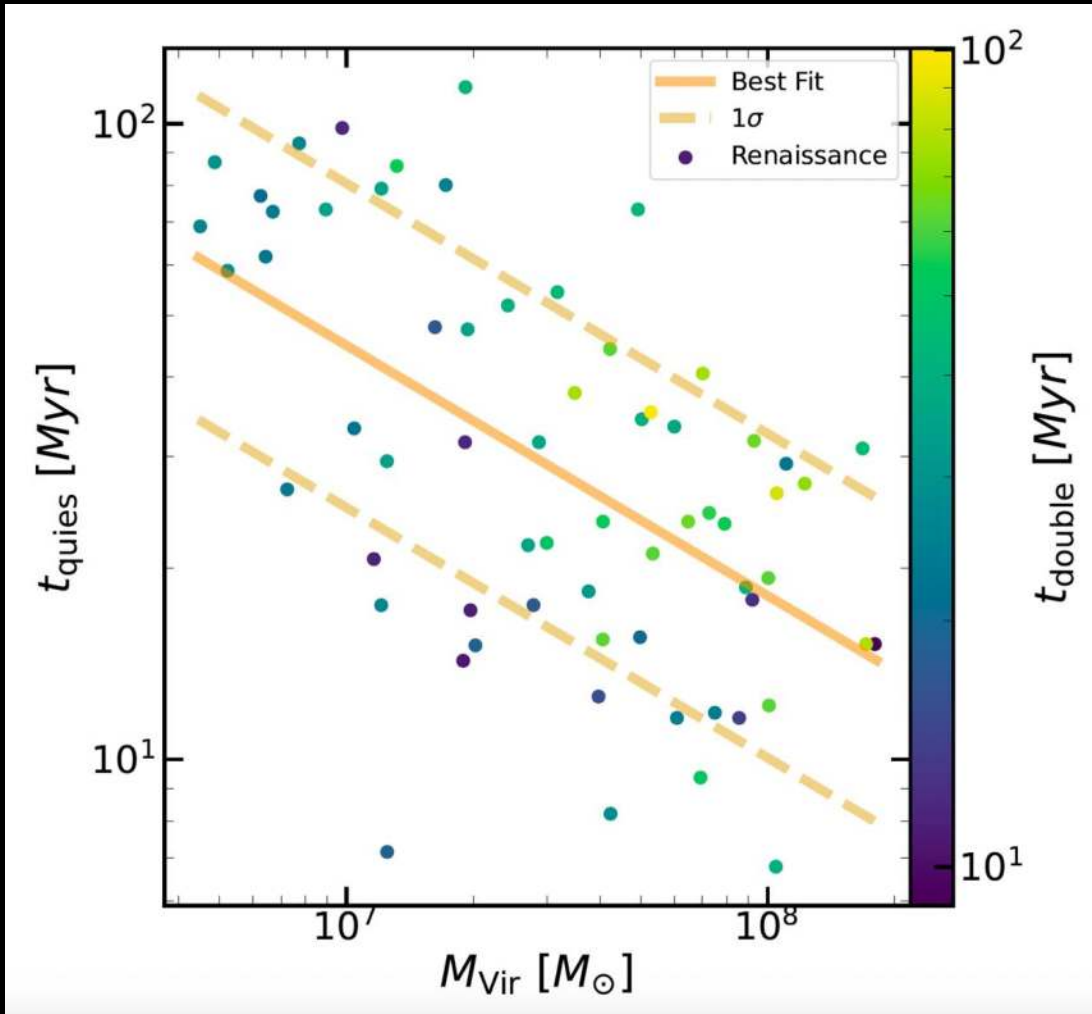
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my Website!

Metal-Enriched Starburst Mass



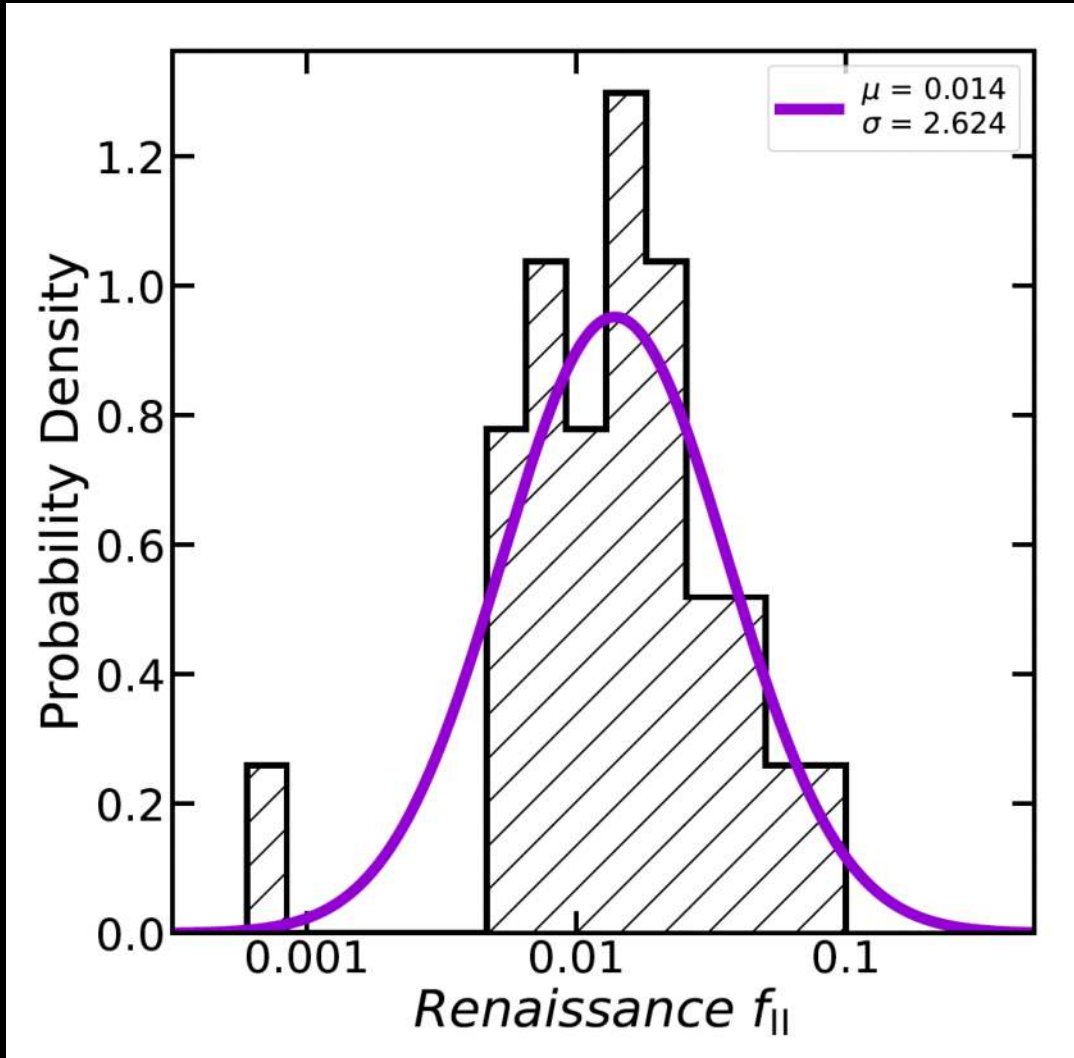
- Each point is from a *Renaissance* star formation history.
- Found best fit power law considering multiple halo properties
- Higher Pop II starburst mass for larger mass halos and at lower redshift

Quiescent Period After Pop II Starburst



- Best fit with quiescent period decreasing for larger mass halos
- Found little correlation with accretion history through t_{double} , not included
- Other halo properties showed weak to no correlation

Steady Phase Star Formation Efficiencies



$$\dot{M}_* = \frac{f_{II}}{t_{ff}} M_{gas}$$

$$\dot{M}_{gas} = \dot{M}_{acc} - \dot{M}_* - \eta_{SN} \dot{M}_*$$

- Pop II formation modeled using coupled differential equations (Furlanetto+2022)
- Get best fit for f_{II} for each steady phase
- Find that Renaissance has weak SN feedback